

```
1 def find_position(matrix, ch):
2     if ch == 'J':
3         ch = 'I'
4
5     for i in range(5):
6         for j in range(5):
7             if matrix[i][j] == ch:
8                 return i, j
9
10
11 def decrypt_playfair(matrix, ciphertext):
12     plaintext = ""
13
14     for i in range(0, len(ciphertext), 2):
15         a = ciphertext[i]
16         b = ciphertext[i + 1]
17
18         r1, c1 = find_position(matrix, a)
19         r2, c2 = find_position(matrix, b)
20
21
22         if r1 == r2:
23             plaintext += matrix[r1][(c1 - 1) % 5]
24             plaintext += matrix[r2][(c2 - 1) % 5]
25
26
```

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27     elif c1 == c2:
28         plaintext += matrix[(r1 - 1) % 5][c1]
29         plaintext += matrix[(r2 - 1) % 5][c2]
30
31
32     else:
33         plaintext += matrix[r1][c2]
34         plaintext += matrix[r2][c1]
35
36     return plaintext
37 matrix = [
38     ['M', 'F', 'H', 'I', 'K'],
39     ['U', 'N', 'O', 'P', 'Q'],
40     ['Z', 'V', 'W', 'X', 'Y'],
41     ['E', 'L', 'A', 'R', 'G'],
42     ['D', 'S', 'T', 'B', 'C']
43 ]
44
45 ciphertext = ""IXJEYUREBEZWEHEWRYTUHEYFS
46 KREHEGOYFIWTTTUOLKSYCAJPO
47 BOTEIZONTXBYBNTGONEYCUZWR
48 GDSONSXBOUYWRHEBAAHYUSEDQ""
49
50 ciphertext = ''.join(ciphertext.split())
51

```

Decrypted Text:

IYMRZQAGDRYVAMAZGXDOMAVKCFAGMAAQVKHXSSDONAFCQGRHONTPDAMXNUBWCXSPCANUGZDQYVARCDNUBVT
PQZXA
MATRWTZQDL CU